Project - Deploy and Secure an Azure VM

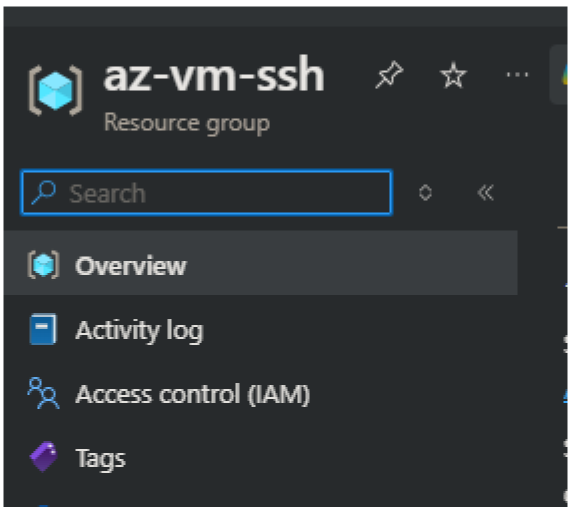
Objective:

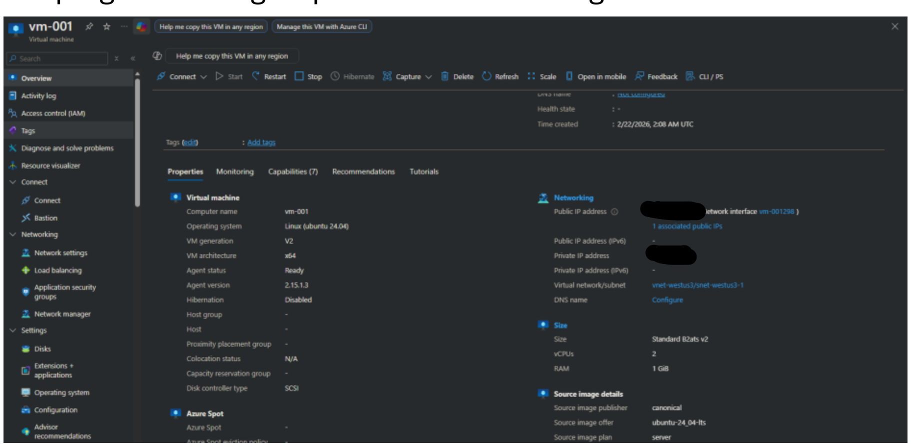
- Create a Basic VM and Secure in Azure

Environment:

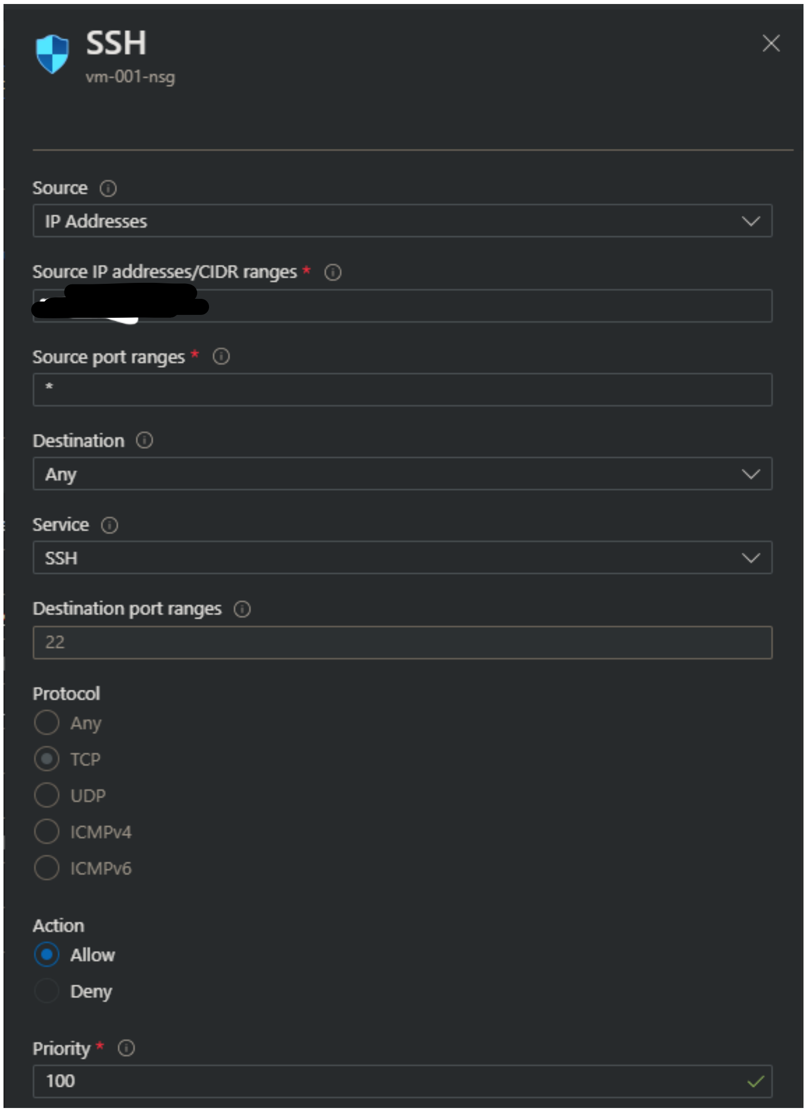
- Azure Free Subscription

Recreation steps:

1. Create a resource group in a subscription
    a. 

2. Once resource is created -> create a VM
    a. This example I used a Linux (ubuntu 24.04)
    b. Keeping resource group and VM in same region
    c. 

3. Next step: Securing NSG to filter traffic to only allow SSH to specific inbound IP (in this case using personal computer to connect only)

   a. 

      i. Using personal computer IP as source will allow only this ip to connect to VM using ssh according to this rule

4. Connecting to VM via SSH
    a. Using personal computer in CMD
        i. We can connect to the VM via command
            1. "ssh -i (key) (azure username) @ (VM public IP)
    b. Once connected we want to secure it
        i.

```
Microsoft Windows [Version 10.0.26100.7628]
(c) Microsoft Corporation. All rights reserved.

C:\Users\123817529>ssh -i "C:\Users\123817529\Downloads\vm-001_key.pem" azureuser@[redacted]
The authenticity of host '[redacted] ([redacted])' can't be established.
ED25519 key fingerprint is SHA256:DwkLxWnJDwqrb1Axz6In9tjPoyBS+UBrOFKlYLukyos.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added [redacted] (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1017-azure x86_64)
```

5. Running basic updates to Linux Server
    a. sudo apt update && sudo apt upgrade –y
        i.

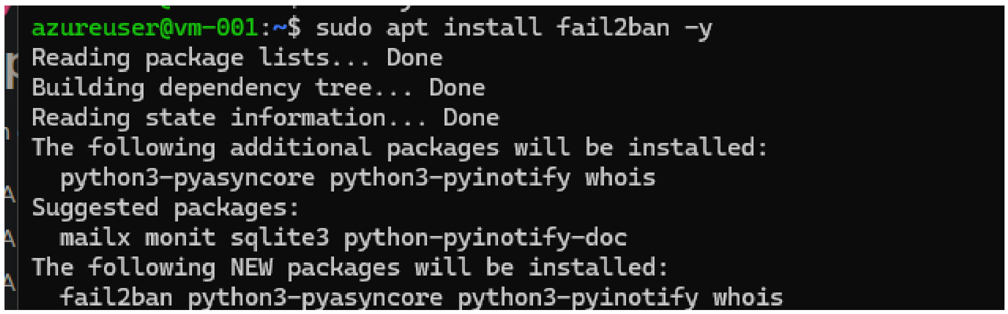
```
azureuser@vm-001:~$ sudo apt update && sudo apt upgrade -y
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:11 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:12 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:13 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1771 kB]
Get:14 http://azure.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [328 kB]
Get:15 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
```

6. Configured a basic firewall UFW
    a. sudo apt install ufw -y
    b. sudo ufw allow ssh
    c. sudo ufw enable
    d. sudo ufw status (check the status of firewall)
        i.

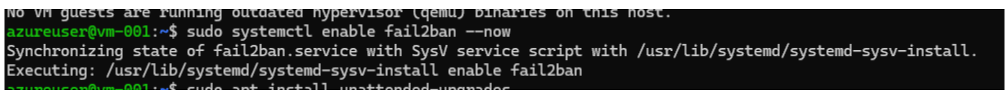
```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
azureuser@vm-001:~$ sudo apt install ufw -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ufw is already the newest version (0.36.2-6).
ufw set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
azureuser@vm-001:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
azureuser@vm-001:~$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
azureuser@vm-001:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)
```

7. Disable password authentication
    a. sudo nano /etc/ssh/sshd_config

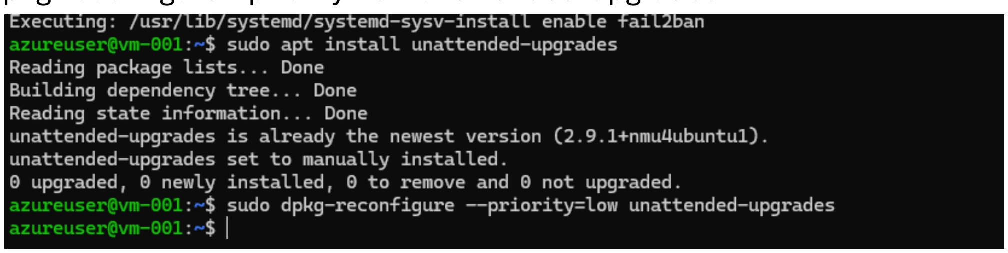
i. In editor change the below to “no”
1. PasswordAuthentication no
2. ChallengeResponseAuthentication no
3. UsePAM yes

8. Restarting SSH
   a. sudo systemctl restart sshd
9. Security hardening : Install fail2ban
   a. sudo apt install fail2ban –y
   b. sudo systemctl enable fail2ban –now
      i. 

```
azureuser@vm-001:~$ sudo apt install fail2ban -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  python3-pyasyncore python3-pyinotify whois
Suggested packages:
  mailx monit sqlite3 python-pyinotify-doc
The following NEW packages will be installed:
  fail2ban python3-pyasyncore python3-pyinotify whois
```



```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
azureuser@vm-001:~$ sudo systemctl enable fail2ban --now
Synchronizing state of fail2ban.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable fail2ban
```

10. Enabling Automatic Security Updates
    a. sudo apt install unattended-upgrades
    b. sudo dpkg-reconfigure --priority=low unattended-upgrades
       i. 

```
Executing: /usr/lib/systemd/systemd-sysv-install enable fail2ban
azureuser@vm-001:~$ sudo apt install unattended-upgrades
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
unattended-upgrades is already the newest version (2.9.1+nmu4ubuntu1).
unattended-upgrades set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
azureuser@vm-001:~$ sudo dpkg-reconfigure --priority=low unattended-upgrades
azureuser@vm-001:~$
```

Summary:

Created Ubuntu VM using SSH key pair

Applied security controls:

OS updates

UFW firewall enabled

SSH locked to key-auth only
Password login disabled
NSG restricted to my IP
Installed fail2ban
Enabled automatic updates